

Regex Symbols Cheat Sheet

Symbol / Pattern	Meaning	Example
<code>^</code>	Start of the string	<code>^abc</code> → String must start with abc
<code>\$</code>	End of the string	<code>abc\$</code> → String must end with abc
<code>.</code>	Any single character	<code>a.c</code> → Matches abc, a1c, a@c
<code>*</code>	Zero or more occurrences	<code>ab*</code> → a, ab, abb, abbb
<code>+</code>	One or more occurrences	<code>ab+</code> → ab, abb, abbb (not a)
<code>?</code>	Zero or one occurrence (optional)	<code>ab?</code> → a or ab
<code>\d</code>	Any digit (0–9)	<code>\d{10}</code> → Exactly 10 digits
<code>\D</code>	Any non-digit character	<code>\D+</code> → abc, @#\$
<code>{n}</code>	Exactly n times	<code>\d{5}</code> → Exactly 5 digits
<code>{n,m}</code>	Between n and m times	<code>{8,12}</code> → 8 to 12 characters
<code>{n,}</code>	At least n times	<code>a{3,}</code> → aaa, aaaa
<code>[A-Za-z]</code>	Any uppercase or lowercase alphabet	Prince
<code>[0-9]</code>	Any digit	12345
<code>[A-Za-z0-9]</code>	Any letter or digit	abc123
<code>[^]</code>	Any character except a space	Used in email validation
<code>\s</code>	Whitespace (space, tab, newline)	" "
<code>\S</code>	Non-whitespace character	abc123
<code>(?=.*\d)</code>	Must contain at least one digit	Password validation
<code>(?=.*[@\$!%*?&])</code>	Must contain at least one special character	Password validation
<code> </code>		OR operator
<code>()</code>	Groups characters together	<code>(abc)+</code>
<code>[]</code>	Character set (match one character from the set)	<code>[abc]</code> → a, b, or c
<code>[^abc]</code>	Match any character except a, b, or c	d, 1, @

The general syntax of using a Regular Expression with test() in an if statement is:

```
if (!regexPattern.test(variable)) {  
    // Code to execute if the pattern does NOT match  
}
```

Syntax Breakdown

```
if (!pattern.test(value)) {  
    // Error message  
    return false;  
}
```

- pattern → Regular expression.
- test() → Checks whether the value matches the pattern.
- ! → Negates the result (true becomes false, false becomes true).
- return false; → Stops form submission if validation fails.

General Template:

```
let pattern = /regular_expression/;  
  
if (!pattern.test(inputValue)) {  
    alert("Error message");  
    return false;  
}
```

document.getElementById() – General Syntax & Common Methods (Revision Notes)

What is document.getElementById()?

- It is a JavaScript DOM method.
- It selects an HTML element using its **id** attribute.
- It returns the specified element so that you can read or modify it.

➤ Syntax

```
document.getElementById("idName");
```

Example:

```
document.getElementById("username");
```

Common Properties and Methods

Method / Property	General Syntax	Purpose
Get Value	document.getElementById("id").value	Get textbox value
Set Value	document.getElementById("id").value = "text"	Set textbox value
Inner HTML	document.getElementById("id").innerHTML	Get HTML content
Change HTML	document.getElementById("id").innerHTML = "Hello"	Change HTML content
Inner Text	document.getElementById("id").innerText	Get plain text
Change Text	document.getElementById("id").innerText = "Welcome"	Change text
Text Content	document.getElementById("id").textContent	Get all text content
Style	document.getElementById("id").style.color = "red"	Change CSS style
Background Color	document.getElementById("id").style.backgroundColor = "yellow"	Change background
Font Size	document.getElementById("id").style.fontSize = "20px"	Change font size
Display	document.getElementById("id").style.display = "none"	Hide element

Show Element	<code>document.getElementById("id").style.display = "block"</code>	Show element
Visibility	<code>document.getElementById("id").style.visibility = "hidden"</code>	Hide without removing space
Enable	<code>document.getElementById("id").disabled = false</code>	Enable input/button
Disable	<code>document.getElementById("id").disabled = true</code>	Disable input/button
Focus	<code>document.getElementById("id").focus()</code>	Place cursor in input
Blur	<code>document.getElementById("id").blur()</code>	Remove cursor focus
Click	<code>document.getElementById("id").click()</code>	Simulate button click
Select	<code>document.getElementById("id").select()</code>	Select textbox content
Reset Form	<code>document.getElementById("formId").reset()</code>	Clear form fields
Submit Form	<code>document.getElementById("formId").submit()</code>	Submit form
Class Name	<code>document.getElementById("id").className</code>	Get/Set class name
Add Class	<code>document.getElementById("id").classList.add("class")</code>	Add CSS class
Remove Class	<code>document.getElementById("id").classList.remove("class")</code>	Remove CSS class
Toggle Class	<code>document.getElementById("id").classList.toggle("class")</code>	Toggle CSS class
Check Checkbox	<code>document.getElementById("id").checked</code>	Get checkbox/radio state
Set Checkbox	<code>document.getElementById("id").checked = true</code>	Check checkbox
Image Source	<code>document.getElementById("id").src = "image.jpg"</code>	Change image
Hyperlink	<code>document.getElementById("id").href</code>	Get/Set link

Most Used in Form Validation

Property	Syntax	Use
Value	<code>.value</code>	Read input value
Length	<code>.value.length</code>	Count characters
Trim	<code>.value.trim()</code>	Remove spaces

Focus	.focus()	Move cursor to field
Checked	.checked	Check radio/checkbox
InnerHTML	.innerHTML	Display error message
Style	.style.color	Change error text color

Example Syntax

- **Read Value**

```
let name = document.getElementById("name").value;
```

- **Set Value**

```
document.getElementById("name").value = "Prince";
```

- **Change Text**

```
document.getElementById("msg").innerHTML = "Login Successful";
```

- **Change Color**

```
document.getElementById("msg").style.color = "green";
```

- **Focus on Textbox**

```
document.getElementById("username").focus();
```

- **Check Checkbox**

```
document.getElementById("agree").checked
```

- **Disable Button**

```
document.getElementById("btn").disabled = true;
```

- **Hide Element**

```
document.getElementById("box").style.display = "none";
```

Easy Way to Remember

- **Email:**

→ Think: Text + @ + Text + . + Text

- **Password**

→ Think: At least 1 digit, At least 1 special character, 8 to 12 characters

Common Date Methods

Method	Description	Example Output
<code>new Date()</code>	Creates current date	Thu Jul 09 2026
<code>new Date(dob)</code>	Converts DOB to Date object	Mon Jan 15 2005
<code>getFullYear()</code>	Returns year	2026
<code>getMonth()</code>	Returns month (0–11)	6 (July)
<code>getDate()</code>	Returns day of month	9
<code>getDay()</code>	Returns weekday (0–6)	4 (Thursday)
<code>getHours()</code>	Returns hour	18
<code>getMinutes()</code>	Returns minutes	30